

Seonghwan Seo

Ph.D. Candidate | Korea Advanced Institute of Science and Technology

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RESEARCH INTERESTS

Generative Modeling

Generative Flow Networks (GFlowNets); diffusion and flow-based models

AI for Drug Discovery

Synthesis-aware molecular design; structure-based drug discovery

Biomolecular Design

Protein design; protein folding and biomolecular co-folding

EXPERIENCE

Intelligent Chemistry Lab, KAIST

AI for Drug Discovery

Supervisor: Woo Youn Kim

Daejeon, South Korea

June 2019 - Present

- Development of an unbiased drug-likeness scoring model
- Development of a deep learning framework for fully automated receptor-based pharmacophore modeling
- Research on synthesis-aware drug discovery
- Development of trillion-scale chemical library search technology with eMolecules & HITS
- National project on K-Fold, a next-generation foundation model for biomolecular co-folding

HITS Inc.

AI Research Intern

Manager: Jaechang Lim

Seoul, South Korea

Dec. 2020 - Aug. 2022

- Development of a reaction template-based generative model for synthesizable molecular design with desired properties
- Development of a transformer-based model for synthesis pathway generation

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST)

Ph.D. student in Chemistry

Supervisor: Woo Youn Kim

Daejeon, South Korea

Aug. 2022 - Feb. 2028 (expected)

Korea Advanced Institute of Science and Technology (KAIST)

B.S. in Chemistry & Computer Science (Double Major)

Supervisor: Young Min Rhee

GPA: 3.99/4.3

Daejeon, South Korea

Feb. 2018 - Aug. 2022

PUBLICATIONS

First author

* indicates equal contribution; † indicates corresponding author

- Boltz2ESI: Accurate Enzyme Specificity Prediction with Co-folding Foundation Model [[link](#)]
Xiwei Cheng*, **Seonghwan Seo***, Jihang Chen, Songlin Jiang, Pengkang Guo, Wengong Jin†
ICLR 2026 Workshop on Generative and Experimental Perspectives for Biomolecular Design (GEM)
- Compositional Flows for 3D Molecule and Synthesis Pathway Co-design [[link](#)]
Tony Shen*, **Seonghwan Seo***, Ross Irwin, Kieran Didi, Simon Olsson, Woo Youn Kim, Martin Ester†
International Conference on Machine Learning (ICML), 2025
- Generative Flows on Synthetic Pathway for Drug Design [[link](#)]
Seonghwan Seo, Minsu Kim, Tony Shen, Martin Ester, Jinkyoo Park, Sungsoo Ahn, Woo Youn Kim†
International Conference on Learning Representations (ICLR), 2025
- PharmacoNet: deep learning-guided pharmacophore modeling for ultra-large-scale virtual screening [[link](#)]
Seonghwan Seo, Woo Youn Kim†
Chemical Science, 2024
- Molecular Generative Model via Retrosynthetically Prepared Chemical Building Block Assembly [[link](#)] [[cover](#)]
Seonghwan Seo, Jaechang Lim&dagger, Woo Youn Kim†
Advanced Science, 2023 (selected for a journal cover)

- Drug-likeness scoring based on unsupervised learning [[link](#)]
Kyunghoon Lee*, Jinho Jang*, **Seonghwan Seo***, Jaechang Lim&dagger, Woo Youn Kim†
Chemical Science, 2022

Co-author

- Atom-level Protein Representation Learning Improves Protein Structure Prediction [[link](#)]
Taewon Kim, Hyosoon Jang, Hyunjin Seo, **Seonghwan Seo**, Hyeonwoo Kim, Wonho Chung, Mingyeong Shin, Wooyoun Kim, Sungsoo Ahn†
arXiv preprint, 2026
- Generalized Multi-State Protein Design with AlphaFold3 [[link](#)]
Xiwei Cheng, Pengkang Guo, **Seonghwan Seo**, Casper A. Goverde, Wengong Jin†
ICLR 2026 Workshop on Generative and Experimental Perspectives for Biomolecular Design (GEM)
- TacoGFN: Target Conditioned GFlowNet for Structure-based Drug Discovery [[link](#)]
Tony Shen, **Seonghwan Seo**, Grayson Lee, Mohit Pandey, Jason R Smith, Artem Cherkasov, Woo Youn Kim, Martin Ester†
Transactions on Machine Learning Research (TMLR), 2024

ACADEMIC SERVICE

- Organizer of the [Generative AI and Biology \(GenBio\)](#), at ICML 2026 in Seoul
- Reviewer (Conferences): ICLR (2025-2026), NeurIPS (2026)
- Reviewer (Journals): Nature Communications (2024)

PRESENTATIONS

Invited

- "Molecular Generative Model via Retrosynthetically Prepared Chemical Building Block Assembly"
Oral workshop and tutorial at *AI-BIO, Artificial Intelligence Institute, Seoul National University*, Seoul, South Korea (Nov. 2022)

Contributed

- "Compositional Flows for 3D Molecule and Synthesis Pathway Co-design"
Oral spotlight & Poster presentation at *ICLR 2025 Workshop on AI for Accelerated Materials Discovery (AI4Mat)*, Singapore. (Apr. 2025)
- "Compositional Flows for 3D Molecule and Synthesis Pathway Co-design"
Oral spotlight & Poster presentation at *ICLR 2025 Workshop on Generative and Experimental Perspectives for Biomolecular Design (GEM)*, Singapore. (Apr. 2025)
- "Generative Flows on Synthetic Pathway for Drug Design"
Poster presentation at *ICLR 2025*, Singapore. (Apr. 2025)
- "Generative Flows on Synthetic Pathway for Drug Design"
Poster presentation at *NeurIPS 2024 Workshop on AI for New Drug Modalities (AIDrugX)*, Vancouver, Canada. (Dec. 2024)
- "PharmacoNet: Accelerating Large-Scale Virtual Screening by Deep Pharmacophore Modeling"
Poster presentation at *NeurIPS 2023 Workshop on New Frontiers of AI for Drug Discovery and Development (AI4D3)*, New Orleans, LA, USA. (Dec. 2023)
- "Molecular Generative Model via Retrosynthetically Prepared Chemical Building Block Assembly"
Poster presentation at *2023 Accelerate Conference*, Toronto, Canada (Aug. 2023)

AWARDS & SCHOLARSHIPS

- **TTA President Award**
Korea AI-Champion Competition, 2025
- **KAIST Alumni Scholarship**
KAIST Alumni Association, 2021 - 2022
- **Younghoon Lee Scholarship**
Department of Chemistry, KAIST, 2021
- **Model Student Awards**
KAIST, 2021
- **Dean's List**
College of Natural Science, KAIST, 2021
- **National Science & Technology Scholarship**

Government of South Korea, 2020 - 2021

- **Dean's List**

School in Freshman, KAIST, 2019

SKILLS

Languages

Korean (native), English (upper-intermediate)

Deep Learning Tools

Python, PyTorch, PyTorch Geometric, PyTorch Lightning, Numba, Pandas

Chemistry Tools

RDKit, OpenBabel, Gemmi, Vina, Smina, rDock